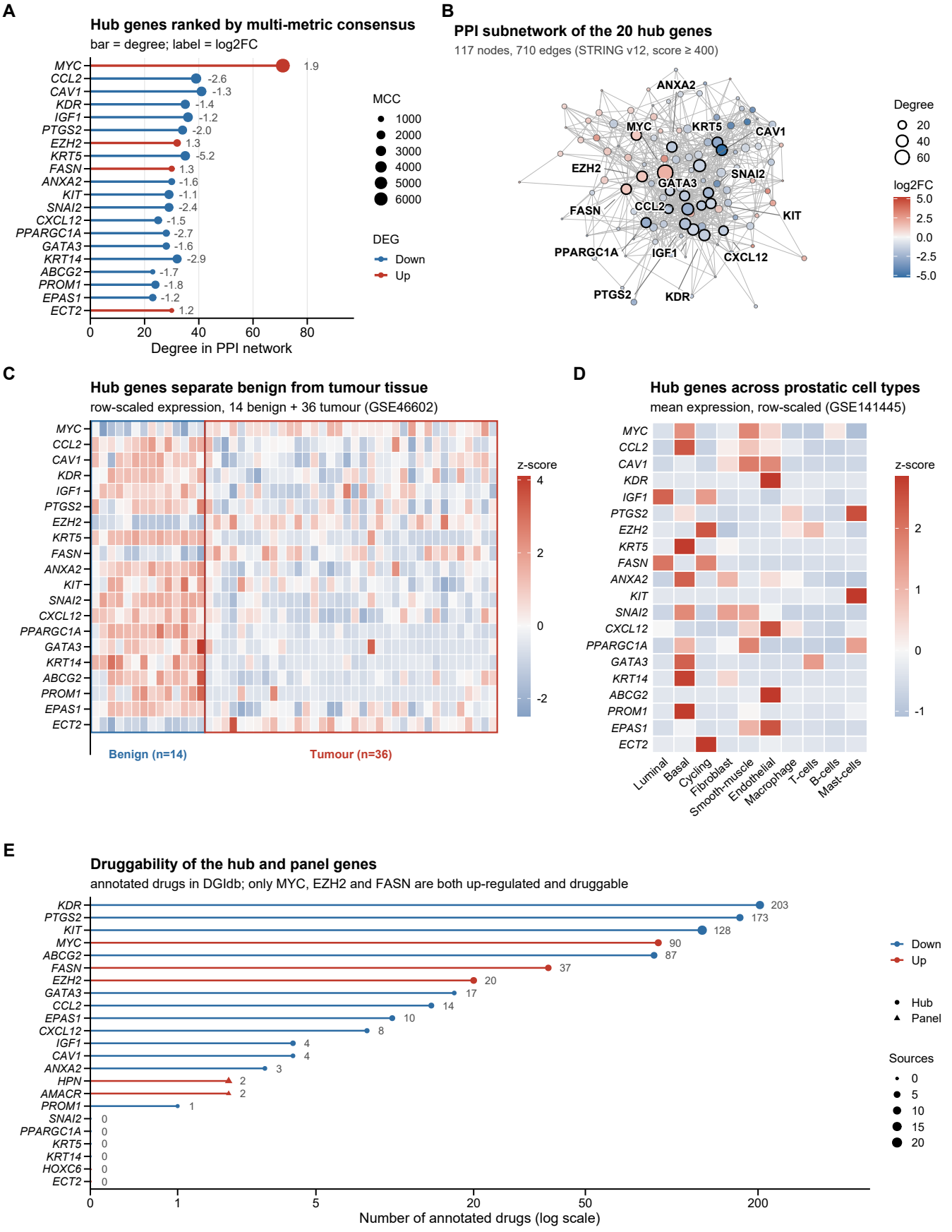

A
Hub genes ranked by multi-metric consensus
bar = degree; label = log2FC
MYC 1.9
CCL2 -2.6
CAV1 -1.3
KDR -1.4
IGF1 -1.2
PTGS2 -2.0
EZH2 1.3
KRT5 -5.2
FASN 1.3
ANXA2 -1.6
KIT -1.1
SNAI2 -2.4
CXCL12 -1.5
PPARGC1A -2.7
GATA3 -1.6
KRT14 -2.9
ABCG2 -1.7
PROM1 -1.8
EPAS1 -1.2
ECT2 1.2
0 20 40 60 80
Degree in PPI network
MCC
1000
2000
3000
4000
5000
6000
DEG
Down
Up
B
PPI subnetwork of the 20 hub genes
117 nodes, 710 edges (STRING v12, score ≥ 400)
ANXA2
MYC
KRT5
CAV1
EZH2
SNAI2
GATA3
FASN
CCL2
KIT
PPARGC1A
IGF1
CXCL12
PTGS2
KDR
Degree
20
40
60
log2FC
5.0
2.5
0.0
-2.5
-5.0
C
Hub genes separate benign from tumour tissue
row-scaled expression, 14 benign + 36 tumour (GSE46602)
MYC
CCL2
CAV1
KDR
IGF1
PTGS2
EZH2
KRT5
FASN
ANXA2
KIT
SNAI2
CXCL12
PPARGC1A
GATA3
KRT14
ABCG2
PROM1
EPAS1
ECT2
Benign (n=14)
Tumour (n=36)
z-score
4
2
0
-2
D
Hub genes across prostatic cell types
mean expression, row-scaled (GSE141445)
MYC
CCL2
CAV1
KDR
IGF1
PTGS2
EZH2
KRT5
FASN
ANXA2
KIT
SNAI2
CXCL12
PPARGC1A
GATA3
KRT14
ABCG2
PROM1
EPAS1
ECT2
Luminal
Basal
Cycling
Fibroblast
Smooth-muscle
Endothelial
Macrophage
T-cells
B-cells
Mast-cells
z-score
2
1
0
-1
E
Druggability of the hub and panel genes
annotated drugs in DGIdb; only MYC, EZH2 and FASN are both up-regulated and druggable
KDR 203
PTGS2 173
KIT 128
MYC 90
ABCG2 87
FASN 37
EZH2 20
GATA3 17
CCL2 14
EPAS1 10
CXCL12 8
IGF1 4
CAV1 4
ANXA2 3
HPN 2
AMACR 2
PROM1 1
SNAI2 0
PPARGC1A 0
KRT5 0
KRT14 0
HOXC6 0
ECT2 0
0 1 5 20 50 200
Number of annotated drugs (log scale)
Down
Up
Hub
Panel
Sources
0
5
10
15
20